

Does a Local Metric Help Anisotropic Alpha-Complex Surface Reconstruction for Additive Manufacturing?

A Frozen Held-Out Evaluation

Abstract

Purpose. In additive-manufacturing (AM) reverse-engineering and scan-to-print workflows a surface mesh is reconstructed from a scanned point cloud, and scale selection for alpha-complex reconstruction governs printed-part fidelity. This paper asks whether a local symmetric-positive-definite (SPD) metric derived from a directed relation-tensor field adds value beyond established density-adaptive and normal-anisotropic alpha baselines.

Design/methodology/approach. A predeclared, frozen held-out protocol with strict information boundaries evaluates the methods on a synthetic six-family panel and a real-mesh corpus, scoring every candidate against a strict casewise per-endpoint baseline envelope on geometry and topology.

Findings. The primary result is negative: on real held-out meshes the normal-anisotropic baseline attains the best mean surface F-score (0.868) and no local-metric variant exceeds it (the closest trails by 0.094, bootstrap 95% CI $[-0.11, -0.08]$, clearing the envelope in 2 of 40 cases). A weighted (regular/power) alpha complex that changes connectivity strictly improves the density baseline on both synthetic and real data, and a deployed fail-closed exact-construction fallback never certifies an unvalidated result.

Practical implications. For AM reconstruction quality assurance a well-regularised normal-based metric remains the strong default; a heavier local metric is not worth

its cost, whereas the weighted-alpha connectivity change and the fail-closed fallback are safe, reusable upgrades.

Originality/value. A rigorous, reproducible held-out comparison that quantifies the (non-)value of a local-metric anisotropic alpha for AM reconstruction, together with a better density baseline, a safety fallback, and a documented universal failure of removal-based false-bridge interventions.

Keywords. Surface reconstruction; Alpha complex; Additive manufacturing; Reverse engineering; Anisotropic metric; Held-out evaluation

Article classification. Research paper

1 Introduction

Reverse-engineering and scan-to-print pipelines in additive manufacturing (AM) begin by reconstructing a surface mesh from a scanned point cloud, and the geometric fidelity and topological soundness of that reconstruction propagate directly into the printed part. A boundary that is over-filled fuses features that should be separate; a boundary that is under-filled opens holes that break watertightness and make the model unsliceable without repair. Because the downstream slicer assumes a clean, orientable, watertight shell, reconstruction error is not cosmetic: it determines whether a scanned part can be re-manufactured at all, and how much manual repair the operator must perform before printing. Reconstruction quality is therefore a genuine AM concern, not only a computational-geometry one.

The alpha complex (Edelsbrunner and Mücke, 1994) is a classical and still widely used tool for this task: from a fixed Delaunay triangulation of the input points it selects a subcomplex governed by a scale parameter α , interpolating between the point set (small α) and the convex hull (large α). Its appeal is that it is parameter-light, provably a subcomplex of the Delaunay triangulation, and comes with a clean topological interpretation. Its weakness is equally well known: a single global scale cannot simultaneously avoid over-fill in dense regions and topology loss in sparse regions when the sampling density varies in

space, as it always does for real scans of parts with mixed feature sizes.

Two families of remedy are established. The first is *density-adaptive* alpha, which normalises each tetrahedron’s circumradius by the local k -nearest-neighbour (kNN) spacing so that the effective scale tracks the sampling density (we denote this baseline B4). The second is *normal-anisotropic* alpha, which replaces the isotropic Euclidean radius with a regularised local principal-component (PCA) SPD metric carrying a planarity-weighted penalty along the estimated surface normal, so that the complex prefers to grow tangentially along the surface rather than across it (B5). Both are strong, cheap, and require only local neighbourhood computations.

Against this backdrop a richer construction has been proposed. From directed kNN scale contrast and receiver-direction imbalance one can build a *signed, trace-free relation tensor* field, map it through bounded log-eigenvalues to a local SPD metric, and score the alpha complex under that metric (we call this candidate P1). A confidence-gated conservative variant falls back to the density baseline on cells where the field is deemed unreliable (P2). Such a field encodes more of the local directional structure than a symmetric normal tensor can, and it is natural to hope that this extra structure sharpens reconstruction where normals alone are ambiguous. Whether that hope survives contact with realistic data is exactly the question a practitioner deciding what to deploy must answer.

Question. Does the added complexity of a local SPD metric provide reconstruction value beyond the existing density (B4) and normal-anisotropic (B5) baselines, under an evaluation strict enough to be trusted?

Contributions. (1) A predeclared frozen held-out protocol with strict information boundaries, applied to a synthetic six-family panel and to a real-mesh corpus, that scores every candidate against a casewise per-endpoint baseline *envelope* rather than a mean. (2) A quantitative, *negative* answer: the local SPD metric does not beat the normal-anisotropic baseline on real held-out data, and a small mean gain on the synthetic panel does not survive the envelope test. (3) A weighted regular (power) alpha complex, M1,

that changes the connectivity itself and strictly improves the density baseline on both synthetic and real data. (4) A deployed fail-closed exact-construction fallback with a no-silent-false-safe guarantee, and a comprehensive negative map showing that removal-based false-bridge interventions fail universally, with an explanation of why.

We stress at the outset that this is an honest negative/limits study. We do not claim to beat prior art; we claim to have measured, carefully, that a specific and intuitively appealing addition does not help, and to have isolated the two pieces that do.

2 Related work

2.1 Alpha shapes and their weighted generalisation

Alpha shapes and the three-dimensional alpha complex (Edelsbrunner and Mücke, 1994) select a subcomplex of the Delaunay triangulation by a single scale, and remain a default reconstruction primitive because of their simplicity and topological guarantees. The weighted (regular, or power) generalisation (Edelsbrunner, 1992) assigns each site a scalar weight and replaces the Delaunay triangulation by the regular triangulation of the weighted points, obtained as a lower convex hull of a lift into one higher dimension. Weights let sites of different importance or density occupy different effective radii; crucially, they change the underlying *connectivity*, not merely the order in which a fixed set of simplices is admitted. Our M1 candidate is a density-driven instance of this classical device, and its ability to remove simplices that a fixed Delaunay triangulation would force to remain is the reason it can do what rescaling a fixed complex cannot.

2.2 Density-adaptive and anisotropic reconstruction

Normalising the alpha scale by local sampling density is a standard way to handle non-uniform clouds and underlies our B4 baseline. Anisotropic reconstruction, in which the notion of “nearby” is stretched by a local metric rather than a scalar radius, is likewise well established; the most common construction estimates a local tangent frame by principal-component analysis and penalises growth along the normal, which presupposes

consistently oriented normals (Hoppe et al., 1992). Our B5 baseline is a regularised, planarity-weighted member of this family. The relation-tensor metric (P1/P2) evaluated here is a more elaborate anisotropy that additionally encodes directed scale contrast; the empirical question is whether that elaboration buys anything over the PCA metric.

2.3 Automatic scale selection

Because a single global α is inadequate, a body of work selects the scale automatically, for example by persistence of topological features across the filtration or by resampling stability of the reconstructed surface. Our B3 baseline instantiates this idea as an unlabelled persistence-plus-resampling-stability selector, and we further probe a resampling-persistence gate as a possible false-bridge detector. As reported below, that gate does not discriminate true from false bridges on our data (discrimination near chance), which is itself informative about the limits of local stability signals.

2.4 Watertight labelling of Delaunay tetrahedra

A distinct and powerful family reconstructs a watertight surface by labelling each Delaunay tetrahedron as inside or outside, typically via a global minimum-cut on the tetrahedral adjacency graph with visibility or normal-consistency terms (Labatut et al., 2009). This global labelling can, in principle, resolve exactly the thin-gap configurations that defeat every local method we test. We do not implement it here—it is a separate, large engineering effort—but we identify it as the one untested avenue that our negative results do not foreclose, and we discuss why it is the natural next step rather than yet another local metric.

2.5 Evaluation methodology

Reconstruction papers frequently report improvements as gains in a mean score over a convenience set of shapes, tuned and evaluated on overlapping data. We deliberately adopt a stricter stance: a predeclared protocol, frozen scales, strict information boundaries between calibration and evaluation, and a casewise per-endpoint baseline envelope that

a candidate must clear *case by case* rather than on average. This methodology is a contribution in its own right, because it is precisely what converts an appealing mean gain (which P2 exhibits on one synthetic profile) into a verdict of no value.

3 Methods

3.1 Problem set-up and evaluation endpoints

Given an observed point cloud sampled from an unknown surface, a reconstruction method returns a triangulated surface. We evaluate each reconstruction against a dense reference on complementary endpoints so that a method cannot trade topology for geometry unnoticed. The *surface F-score* is the harmonic mean of precision and recall of reconstructed versus reference surface samples under a fixed distance tolerance (higher is better) and is our headline geometric-completeness measure. The *geometric loss* is a normalised two-sided (Chamfer-type) surface distance between reconstruction and reference (lower is better). Topological soundness is summarised by the *Betti error*, the absolute discrepancy in Betti numbers against the ground-truth shell, and by the count of *nonmanifold edges* (edges incident to other than two triangles), both of which a printable shell should drive to zero. Finally, we count *false-bridge edges*: edges that connect points belonging to distinct ground-truth surface sheets, i.e. spurious connectivity that fuses features which should stay separate. A method is credible only if it improves the F-score without regressing topology, which is why our acceptance test is casewise and multi-endpoint rather than a single averaged number.

3.2 Baselines and candidates

B0–B3 are, respectively, a convex hull; a fixed normalised-radius alpha; an exhaustive critical-scale oracle allowed to consult the dense reference (an upper bound, not a deployable method); and an unlabelled persistence-plus-resampling- stability scale selector. B4 divides each tetrahedron’s Euclidean circumradius by the geometric-mean kNN spacing of its vertices, giving density adaptation. B5 combines a regularised local PCA SPD met-

ric with the same density normalisation and a planarity-weighted normal penalty, giving normal anisotropy. P1 forms the relation-tensor SPD metric described next; P2 adds a $\max(\text{P1}, \text{B4})$ guard on low-confidence cells so that it can never score below the density baseline.

Critically, B4, B5, P1 and P2 all select subcomplexes of *one fixed isotropic Delaunay triangulation* by a per-cell scalar threshold. They differ only in how they score a cell, never in which cells exist. It follows that a false-bridge tetrahedron already present in that triangulation cannot be removed by any of them without also removing legitimate incident simplices—a structural ceiling that motivates M1.

3.3 The relation-tensor local metric (P1/P2)

For each site the field construction compares directed kNN scale contrast against its neighbours and measures the imbalance of receiver directions, yielding a signed, trace-free relation tensor $\mathbf{T}$ at that site. The antisymmetric and deviatoric structure of $\mathbf{T}$ encodes directional relationships that a symmetric normal tensor cannot represent. To obtain a usable metric, $\mathbf{T}$ is symmetrised and passed through bounded log-eigenvalues, $\mathbf{M} = \exp(\text{clip}(\log\text{-spectrum}))$, which guarantees a well-conditioned SPD metric with eigenvalues confined to a fixed band, preventing degenerate stretches. Cells are then scored by their circumradius measured under $\mathbf{M}$ instead of the Euclidean radius. P2 additionally computes a per-cell confidence from the field and, where confidence is low, falls back to the B4 score, so P2 is by construction a conservative interpolation between P1 and the density baseline.

3.4 M1: weighted regular (power) alpha complex

M1 departs from the fixed-triangulation family by changing the connectivity itself. Each point receives a density weight $w_i = (s \cdot \text{spacing}_i)^2$, where spacing_i is its local kNN spacing and $s \geq 0$ is a single global weight scale. The regular (weighted) triangulation is the lower convex hull of the lift

$$p_i \longmapsto (p_i, \|p_i\|^2 - w_i), \quad (1)$$

so that with $w \equiv 0$ the construction reproduces the ordinary Delaunay triangulation and M1 strictly generalises B4. Cells are scored by the *proper* weighted (power) circumradius normalised by kNN spacing. This detail is not cosmetic: scoring the weighted connectivity with the *ordinary* circumradius reintroduces nonmanifold edges and Betti regressions, whereas the power circumradius—the natural radius in the weighted setting—removes them, so the two design choices (weighted connectivity, power scoring) must be made together. Because a sufficiently large weight submerges a point beneath the lower hull and deletes it from the triangulation, the weight scale s is capped; we freeze s on calibration data.

3.5 G4: deployed fail-closed exact fallback

Floating-point Delaunay construction can, on adversarial inputs, produce a topologically invalid complex. For a deployment that must never silently certify a bad result, we attach an exact Euclidean Delaunay backend that runs under host validation and, on acceptance, injects its validated connectivity into the actual selection path. On any refusal—a point-count cap exceeded, an exact cospherical ambiguity, host rejection, or backend timeout or malformed output—the method *fails closed* to a floating construction that is explicitly labelled non-exact, and records the specific reason. The *certified* outcome is reachable only after full validation passes; every failure mode records a reason and is never silently treated as safe. A suite of nine failure-mode tests exercises each refusal path, and the default 96-point evaluation panel always fails closed under the deliberately conservative 64-point cap. The guarantee is deliberately narrow: G4 certifies only the base Delaunay connectivity, not the anisotropic complex built on top of it.

3.6 Frozen held-out protocol and information boundaries

Every method’s scale, and M1’s weight scale s , is frozen on a *calibration* set; no tuning of any kind is permitted on the held-out set. Reference clouds and ground-truth labels enter calibration and final scoring only—never held-out selection—so that no method can peek at the answer while choosing its complex. Each candidate is judged against the

strict casewise per-endpoint baseline *envelope* $\max(B4, B5)$: to clear the gate on a case it must match or beat, endpoint by endpoint, whichever baseline is stronger *on that case*. This is a far harder bar than beating either baseline on average, and it is what prevents a candidate from claiming victory by winning easy cases while losing hard ones.

4 Experiments

4.1 Synthetic six-family panel

The synthetic panel comprises six shape families chosen to exercise distinct failure modes—disconnected parts, closely opposing sheets, thin tubes, and mixed-density regions among them. Scales are frozen on a calibration instance of the six families, and evaluation runs four held-out profiles (a base profile plus sparser, noisier, and harder-geometry variants) over three paired random seeds, so that every method sees identical clouds. The panel is intentionally adversarial and, as we show, contains one pathology (an inter-sheet gap smaller than the sample spacing) that no local method can resolve.

4.2 Real-mesh corpus and splits

The real corpus draws watertight meshes from a public 3D-printing model set (Zhou and Jacobson, 2016) together with CAD primitives. Observed clouds are area-weighted surface samples with additive noise, and dense reference clouds are drawn from the same meshes. Scales are frozen on a calibration mesh set and evaluated on a disjoint held-out mesh set across density and noise splits, for 40 held-out cases in total. This is the setting that matters for the AM question, because it is where the baselines’ assumptions (well-defined normals on smooth shells) actually hold.

Table I. Synthetic calibration, frozen multipliers. M1 ($s=0.375$) strictly dominates the density baseline B4 on every endpoint. B5’s topology figures reflect a panel-specific thin-gap pathology, not its behaviour on real meshes.

Method	F-score	Geom. loss	Betti err.	Bridge edges	Nonmanifold
B4	0.365	0.1802	9	48	0
M1 ($s=0.375$)	0.384	0.1730	9	46	0
B5	0.391	0.1824	105	61	175

5 Results

5.1 Synthetic preflight

The frozen confidence-fallback variant P2 preserves the density guard (zero violations) but fails the strict casewise envelope on all four held-out profiles, with mean F-score margins -0.048 , $+0.0002$, -0.049 and -0.049 . The one near-zero, slightly positive mean margin is instructive: it is exactly the kind of small averaged gain that a weaker evaluation would report as an improvement, yet under the casewise envelope it is not evidence of value. On calibration, M1 at weight scale $s = 0.375$ strictly dominates B4 on every endpoint (Table I)—higher F-score, lower geometric loss, no Betti or nonmanifold regression, and fewer false-bridge edges—whereas the normal-anisotropic baseline B5 is topologically degenerate on this panel (Betti error 105, 175 nonmanifold edges), an artefact of the panel’s thin-gap pathology rather than a property of B5 in general. On the held-out envelope M1 is 4–5× closer than the previous best (per-profile F-margins $-0.0105/-0.0005/-0.0003/-0.0109$) but does not clear it; the residual gap is dominated by one family’s false bridges, which we dissect below.

5.2 Real held-out data (the headline)

Table II is the central result. On real meshes the normal-anisotropic baseline B5 is clearly the strongest (mean F-score 0.868, mean geometric loss 0.058), and no local-metric candidate exceeds it. The closest candidate, M1, trails by 0.094 with a bootstrap 95% confidence interval that excludes zero, and clears the casewise envelope in only 2 of 40 cases; the relation-tensor candidates P1 and P2 trail by 0.14 and clear it in 0 of

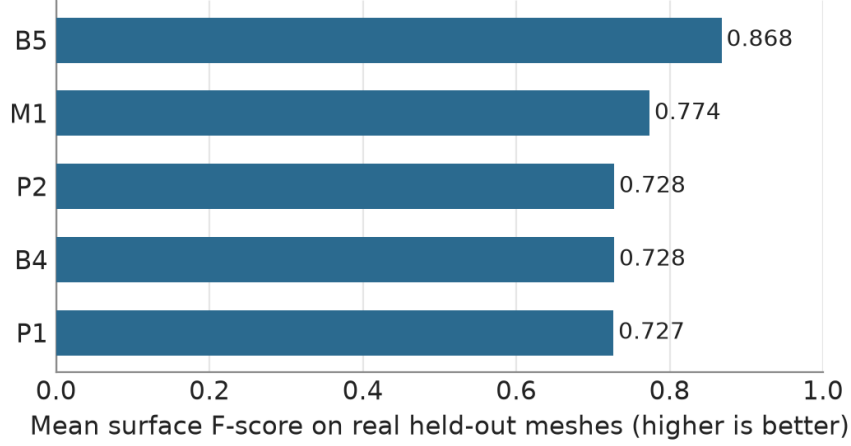


Figure 1. Mean surface F-score on the real held-out corpus. The normal-anisotropic baseline (B5) is best; the weighted regular alpha (M1) improves the density baseline (B4) but no local-metric candidate (P1, P2, M1) reaches B5.

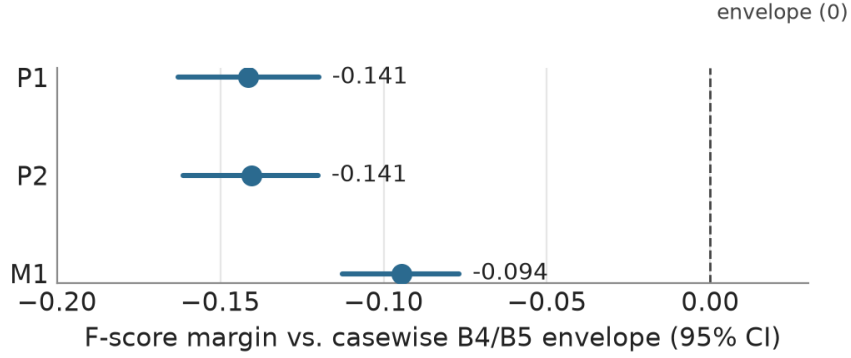


Figure 2. F-score margin of each candidate against the casewise B4/B5 envelope, with 95% bootstrap confidence intervals. Every interval is below the zero envelope, so no candidate clears the held-out gate.

40. Figure 1 shows the ranking and Figure 2 the candidate margins against the baseline envelope with 95% confidence intervals; every candidate interval lies entirely below zero. This reverses the synthetic picture: on smooth real meshes the well-defined normals make B5 excel, while its synthetic degeneracy was an artefact of a panel pathology rather than a real weakness. The practical reading is direct—the local SPD metric adds no value on real data, and the honest default remains a well-regularised normal-based metric.

M1 nonetheless keeps its secondary win: it beats the density baseline B4 on real data too (0.774 vs. 0.728), confirming that the connectivity change transfers from synthetic to real, even though it does not close the gap to B5.

Table II. Real held-out corpus (40 cases): mean performance and margin against the casewise baseline envelope $\max(B4, B5)$.

Method	Mean F-score	Mean geom. loss	F-margin vs. $\max(B4, B5)$
B4	0.728	0.094	—
B5	0.868	0.058	—
P1	0.727	0.092	−0.14 (0/40 cleared)
P2	0.728	0.092	−0.14 (0/40)
M1	0.774	0.078	−0.094, CI [−0.11, −0.08] (2/40)

5.3 A negative map of false-bridge interventions

Because false bridges dominate the residual gap, we tried to remove them directly, and report that every removal-based intervention fails. Per-cell penalties, layered boundary-owner peeling, connected risk-region removal, a resampling-persistence gate (discrimination $AUC \approx 0.50$, i.e. chance), global multiplier backoff, reference-free fail-closed routing, and oriented-normal removal, offset and signed labelling all regress geometry or topology rather than improving them. Two causes explain the universality of the failure. First, removing a cell from a *fixed* complex exposes its interior faces as new boundary, so deletion trades a false bridge for a fresh hole. Second, in the hardest family the inter-surface gap is smaller than the sample spacing—41% of kNN edges cross the two ground-truth sheets—so no *local* method has the information to separate them at all. Only the connectivity change (M1) helps, and even it does not resolve that under-resolved case, because the missing information is genuinely absent from local neighbourhoods.

6 Discussion

The real held-out evaluation shows, quantitatively and under a deliberately strict test, that the local SPD metric (P1/P2) provides no reconstruction value beyond the established normal-anisotropic baseline. The synthetic panel misleads in the opposite direction: by containing an under-resolved thin-gap pathology it unfairly weakens B5, so a synthetic-only study could have reported a spurious win for anisotropy. The divergence between the two settings is itself the lesson—reconstruction claims that rest on adversarial synthetic panels can invert on realistic meshes, and only a real held-out test settles the matter.

For AM practice the implication is reassuring and specific. A well-regularised normal-based metric remains the strong default for scan-to-print reconstruction; the extra machinery of a directed relation tensor is not worth its cost. What is worth adopting are the two pieces that survived: the weighted-alpha connectivity change (M1), which improves the density baseline cheaply and by a mechanism—editing which simplices exist—that rescoring a fixed complex cannot emulate; and the fail-closed exact fallback (G4), which lets an exact construction be deployed on the actual selection path with a guarantee that no failure mode is silently certified as safe, a property that matters when reconstruction feeds an automated print queue.

Methodologically, the study argues for casewise, multi-endpoint, frozen held-out evaluation as the default for reconstruction claims. It was exactly this protocol that demoted P2’s small positive mean margin to non-evidence, and it is exactly the kind of protocol that would prevent the synthetic panel from being over-read.

7 Limitations and future work

The evaluation uses a locally assembled corpus rather than a licensed public leaderboard, and topology endpoints on complex shells are inherently noisy, so the Betti and nonmanifold figures should be read as directional rather than exact. The synthetic panel’s hardest family is under-resolved by construction and cannot be solved by any local method, which bounds what any local candidate could have achieved there.

The one avenue our negative results do not foreclose is global: inside/outside labelling of Delaunay tetrahedra by a minimum-cut over the tetrahedral adjacency graph (Labatut et al., 2009). Unlike every method tested here, such a labelling has, in principle, access to the global evidence needed to close a thin gap without opening a hole, precisely because it does not decide cell by cell. Implementing and held-out testing that labelling—under the same frozen protocol used here—is the natural next step, and a more promising one than any further local metric.

8 Conclusion

We asked whether a PFTF-derived local SPD metric improves anisotropic alpha-complex surface reconstruction for additive manufacturing, and answered it with a predeclared, frozen held-out evaluation on synthetic and real data. The answer is negative: on real held-out meshes a normal-anisotropic baseline is best and no local-metric variant surpasses it. The study nonetheless yields three reusable results—an improved density baseline via a weighted regular alpha (M1), a fail-closed exact-construction fallback with a no-silent-false-safe guarantee (G4), and a documented, mechanistically explained universal failure of removal-based bridge interventions—and it makes the case for casewise frozen held-out evaluation as the standard by which such reconstruction claims should be judged.

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